

STUDENT ACADEMIC PERFORMANCE REPORT AND COURSE SYLLABUS

Student Information

Particulars	Details
Student Name	Gautham Manoj
Roll Number	AM.ID.R4IDS23007-FT
Academic Program & Branch	Ph.D. 2023
Current CGPA	8.76

Semester Performance Details

Semester 3

Semester	Course Code	Course Name	Academic Term Period	Course Type	Grade Obtained
3	SS855	Research Methodology in Applied Sciences	2024-2025 Odd Semester	Regular	B+

Performance Summary

Metric	Score
SGPA	8.0
CGPA	8.76

Registrar's Note

The CGPA value will be displayed only after all courses registered by the student in the respective term have been published by the Registrar's Office.

Course Details

Course Code: SS855

Course Title: Research Methodology in Applied Sciences

Grade Obtained: B+

COURSE SYLLABUS

UNIT I – INTRODUCTION

Research Methodology: Definition and significance of research methodology, objectives of research, types of research, steps in research, identification, selection and formulation of research problem.

UNIT II – RESEARCH FORMULATION

Research Formulation: Problem definition, research design, research plan preparation, hypothesis, review of literature, internet as a source in identifying gap areas from literature reviews and emerging trends, role of computers in research.

Sampling Techniques: Types of sampling, steps in sampling, sample size, advantages and limitations of sampling.

Data for Research: Primary data, collection methods, observation, interview, questionnaire, experimental and case studies, secondary data, relevance, limitations and cautions.

Processing Data: Checking, editing, coding, transcriptions and tabulation.

UNIT III – OVERVIEW OF METHODS IN STATISTICS

Overview of Methods in Statistics: Sampling methods, classification and tabulation of data, sampling design, data processing and analysis strategies, numerical and graphical data analysis.

Data Analysis: Data analysis with statistical packages (SPSS and overview of available software packages).

Hypothesis Testing and Statistical Techniques: Hypothesis testing, generalization and interpretation, ANOVA, t-test and multivariate analysis techniques.

UNIT IV – RESEARCH REPORT

Research Report: Types of reports, styles of reporting, drafting reports, editing and evaluating the final draft.

Research Proposal and Team Work: Developing a proposal and working in a research team.

Critical Appraisal of Published Research: Guidelines for appraisal of published research and steps in the preparation of report.

Report Writing and Documentation: Layout, structure and language of typical reports, illustrations and tables, bibliography, referencing and footnotes.

Oral Presentation: Planning, preparation, practice, making presentation, use of visual aids and importance of effective communication.

UNIT V – ETHICS IN RESEARCH

Ethics in Research: Ethical issues in research.

Intellectual Property Rights (IPR): Copyright, royalty, patents and geographical indications.

Use of Published Material: Reproduction of published material, citation and acknowledgement.

Academic Integrity: Reproducibility and accountability, policy on academic honesty and integrity, plagiarism and unscientific practices in thesis work.

Ethics in Science and Research Funding: Ethics in science and opportunities to carry out research projects with funding or assistance from various Government agencies.

Course Completion Summary

Course Code	Course Title	Semester	Grade	SGPA	CGPA
SS855	Research Methodology in Applied Sciences	3	B+	8.0	8.76

Prepared as a consolidated academic record containing student performance details and complete course syllabus for SS855 – Research Methodology in Applied Sciences.